Summary in Graph

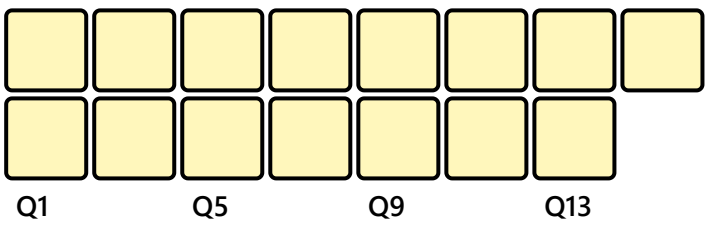
Exam Summary (GO Classes CS Test Series 2026 | Discrete Mathematics | Topic Wise Test 2)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (115) out of 1148 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Select Type Award: 1 Penalty: 0 Set Theory & Algebra

For each of the given (input, output) relationships in Tables One and Two; answer the following:

Table One		Table Two	
A	B	C	D
0	2	1	3
1	3	2	6
2	3	3	9
3	4	4	12

Which of the following is true?

A. A a function of B
B. B a function of A
C. C a function of D
D. D a function of C

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 05sec | Discuss

Q #2 Multiple Select Type | Award: 1 | Penalty: 0 | Set Theory & Algebra

$\{2, 4\}$ is an element of the following sets:

A. $\{\{\{2, 4\}\}\}$
B. $\{1, 2, 3, 4, \{2, 4\}\}$
C. $\{\{1\}, \{4\}, \{2, 4\}\}$
D. $\{(x, x^2) \mid x \in \mathbb{Z}\}$

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

Let A and B be arbitrary sets and consider the set S defined below:

$$S = \{x \mid \neg(x \in A \rightarrow x \in B)\}$$

What is the correct expression for S in terms of A and B using the standard set operators?

A. $A \cup B$
B. $A \cap B$
C. $A \backslash B$
D. $B \backslash A$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 Numerical Type | Award: 1 | Penalty: 0 | Set Theory & Algebra

Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{2, 3, 4\}$. How many sets C have the property that $C \subseteq A$ and $B \subseteq C$?

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 Multiple Select Type | Award: 1 | Penalty: 0 | Set Theory & Algebra

Suppose that S and T are sets. Which of the following first-order logic statements are translations of the statement " S is not a subset of T ?" Check all that apply.

A. $\forall x.(x \in S \rightarrow x \notin T)$
B. $\forall x.(x \in S \wedge x \notin T)$
C. $\exists x.(x \in S \rightarrow x \notin T)$
D. $\exists x.(x \in S \wedge x \notin T)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 Numerical Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Let $\mathrm{A}=\{\mathrm{a}, \mathrm{b}, \mathrm{c}\}$. How many different binary relations are there over A that are partial orders?

Your Answer: | Correct Answer: 19 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Let R_1 and R_2 be relations on a set A.

Consider the following statements:

M: if R_1 and R_2 are both reflexive, then so is $R_1 \cup R_2$.

N: if R_1 and R_2 are both symmetric, then so is $R_1 \cup R_2$.

Which of the above statements is/are true?

A. Only M
B. Only N
C. Both
D. None

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Let f be a function from a set X to a set Y. Consider the following statements.

- P : For each $x \in X$, there exists $y \in Y$ such that $f(x)=y$.
- Q : For each $y \in Y$, there exists $x \in X$ such that $f(x)=y$.
- R : There exist $x_1, x_2 \in X$ such that $x_1 \neq x_2$ and $f\left(x_1\right)=f\left(x_2\right)$.

The negation of the statement " f is one-to-one and onto Y " is

A. P or not R
B. R or not P
C. R or not Q
D. P and not R

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Let's suppose that we have a function $f : \mathbb{R} \to \mathbb{R}$. We'll say that f is an odd function if the following is true:

$$\forall x \in \mathbb{R}.\ f(-x) = -f(x)$$

We can define even functions as follows. A function $f : \mathbb{R} \to \mathbb{R}$ is called even if the following is true:

$$\forall x \in \mathbb{R}.\ f(-x) = f(x)$$

Which of the following statements is/are correct?

A. If $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ are odd, then $g \circ f$ is also odd.
B. If $f : \mathbb{R} \to \mathbb{R}$ is odd and is a bijection, then f^{-1} is also odd.
C. If $f : \mathbb{R} \to \mathbb{R}$ is an even function, then f is not a bijection.
D. Let $f : \mathbb{R} \to \mathbb{R}$ be any function. Then $g : \mathbb{R} \to \mathbb{R}$ defined as $g(x) = f(x) - f(-x)$ is odd.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Let R be an equivalence relation on a non-empty set A. Let $a, b \in \mathrm{A}$.

For any $x \in A$, let $[x]$ denote the equivalence class of R containing x.

Which of the following is/are correct statements?

A. $a\mathrm{R}b$ iff $[a] = [b]$
B. $a\mathrm{R}b$ iff $[a] \cap [b] \neq \phi$
C. If A is non-empty finite set then R can never have more than $|\mathrm{A}|$ equivalence classes, where $|\mathrm{A}|$ is the cardinality of A.
D. $\mathrm{R} = \mathrm{A} \times \mathrm{A}$ iff R has only one equivalence class.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Numerical Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Let A, B be two non-empty sets, with cardinality $3, 4$ respectively. Let R be a relation defined on the power set of $\mathrm{A} \times \mathrm{B}$. Relation R is reflexive, symmetric, transitive, and antisymmetric. Hence, Relation R is also equivalence relation. The cardinality of the largest equivalence class of relation R is?

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

For any set X, let $|\mathrm{X}|$ be the cardinality of set X. Let A and B be two finite sets such that $|\mathrm{A}| = |\mathrm{B}|$. There is a one to one function f from A to B.

Which of the following must be true for f?

- $\mathrm{S1}$: f is onto function
- $\mathrm{S2}$: f has an inverse

A. Only $\mathrm{S1}$

B. Only $\mathrm{S2}$
C. Both $\mathrm{S1}$ and $\mathrm{S2}$
D. None of the above

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Numerical Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Given a set of values $\mathrm{R} = \{1, 2, 3, 4, 5, 6, 7\}$. The number of relations on this set which are both partial-order and equivalence relation is?

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

A binary relation R on a set A is called connected iff for all elements x and y of A, either $x\mathrm{R}y$ or $y\mathrm{R}x$ or both. i.e. R is connected iff $\forall x, y \in \mathrm{A}((x\mathrm{R}y) \vee (y\mathrm{R}x))$

Which of the following is/are true ?

A. Every connected relation is reflexive.
B. Every connected relation is partial order.
C. Every connected relation is symmetric.
D. Every connected relation is transitive.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Let R be a relation from a set A to a set B. The inverse relation from B to A, denoted by R^{-1}, is the set of ordered pairs $\{(b, a) \mid (a, b) \in \mathrm{R}\}$.

- $\mathrm{S1}$: R is reflexive relation iff $\mathrm{R}^{-1} = \mathrm{R}$
- $\mathrm{S2}$: R is a symmetric relation iff $\mathrm{R}^{-1} = \mathrm{R}$

Which one of the following statements is true?

A. Only $\mathrm{S1}$
B. Only $\mathrm{S2}$
C. Both $\mathrm{S1}$ and $\mathrm{S2}$
D. None of the above

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

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